

Fig. 8: IEEE 802.15.4 PHY MAC

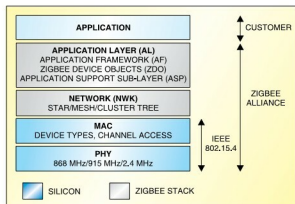


Fig. 9: ZigBee protocol stack

new nodes are sometimes integrated into them during their lifecycle. The ZigBee coordinator must be able to establish links with these new nodes at any point in the network and assign their roles either as FFD or RFD. Hence, the ZigBee coordinator has to store a map of the entire network, identify which nodes within the network are FFDs or RFDs and what are their functions. The ZigBee coordinator therefore requires more local program memory and the ability to address that memory.

Although a network coordinator uses different controllers from those used by other network devices, its development tool should be consistent with them. Future maintenance and upgrades should also be considered during its development. It is therefore imperative that the engineers consider the following factors when choosing a controller for the ZigBee network coordinator:

1. Performance
2. On-chip resources

3. Development tools
4. Compatibility and upgradability
5. Providers

The MCF5208 microprocessor, for instance, matches the need of a ZigBee coordinator. It provides low cost and flexible platform that can be adapted to meet the requirements of even the most complex ZigBee-based systems.

ZigBee router. The ZigBee router (ZR), or lower-level FFD, extends network area coverage, dynamically routes around obstacles and provides back-up routes in case of network congestion or device failure. It can connect to the coordinator and other routers.

ZigBee end device. ZED or RFD can send or receive messages but cannot forward them upstream or downstream. It must be connected to either a coordinator or a router.

The ZigBee standard therefore aims at the following goals:

1. 250-kbps over-the-air data rates
2. Star, tree or peer-to-peer operation
3. Guaranteed time slots (GTSs) and beaconing for communication
4. CSMA-CA channel access
5. Acknowledged messaging for reliable data transfer
6. Low power
7. Short-range operation
8. Reasonable battery life
9. Simple and flexible protocol

As shown in Fig. 9, the IEEE 802.15.4 standard shows the OSI upper layers collapsed into two and provides two services to the upper levels—the PHY data and management services. PHY's features include radio transceiver activation/deactivation, radio channel selection, energy level detection (ED) and received signal quality (LQI), clear channel assessment (CCA), and transmitting and receiving packets across the 2.4-2.48 GHz ISM frequency band.

Layer two is divided into MAC and LLC sub-layer. MAC generically determines who is allowed access to the physical medium at a given time. Its services include beacon management, channel access, GTS management, frame validation, acknowledged frame delivery, association and disassociation, and security mechanisms. LLC sits above the MAC, multiplexes the protocols transmitted over the MAC and provides optional flow control.

The ZigBee stack is divided into network and application layers. The network layer provides services for network formation, security, routing discovery, maintenance and address assignment. The application layer maintains binding tables, defines ZDO and supports manufacturer-defined application objects, whereas the user application builds on the stack through its application interface.

ZigBee commissioning. While commissioning any ZigBee network, challenges like physical attribution of nodes, quantity of nodes, or RF interference, must be taken care of.

ZigBee compliance. ZigBee defines several levels of compliance, one being the ZigBee Compliant Platform (ZCP), which is applied to modules or platforms that are meant for end products. Another compliance level is called ZigBee Network Capable (ZNC), which addresses end products that are built on a ZCP and use a non-public profile. The final compliance level is ZigBee Certified Product (ZCP), which applies to end products that use a public profile.

The world is currently witnessing an enormous growth in the use of WSNs for monitoring and control in the consumer, commercial and industrial sectors, with applications in building automation, industrial control, security applications, home automation, etc. ●

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